

PROJECT MODULE: TASK 5 CONFIGURATION REPORT

Target Infrastructure: Ubuntu 24.04 LTS Instance on Microsoft Azure

1. Scope & System Objective

Deploy a transport-layer packet filtering boundary to reject default unauthorized inbound connections, while keeping communication lines open for explicitly declared destination interfaces.

2. Step-by-Step Implementation Sequence

1. **Default Security Baseline Modification** : Configured Uncomplicated Firewall (UFW) internal rules logic to implement a default deny structure:

Bash

```
sudo ufw default deny incoming
```

```
sudo ufw default allow outgoing
```

2. **Ingress Control Rules Insertion** : Injected precise management parameters boundaries matching assignment criteria:

Bash

Restrict open management control planes connections exclusively to verified client IP coordinates

```
sudo ufw allow from <MY_LOCAL_LAPTOP_IP_HERE> to any port 22 proto tcp
```

Permit standard public transport interface layers

```
sudo ufw allow 80/tcp
```

```
sudo ufw allow 8000/tcp
```

3. **Rule Enforcement Trigger** : Activated the host filtering subsystem to lock active rules states immediately:

Bash

```
sudo ufw enable
```

3. Verification & Validation Metrics

Auditing the active state layout matrix prints the deployed host firewall policies cleanly:

Plaintext

```
$ sudo ufw status verbose
```

Status: active

Logging: on (low)

Default: deny (incoming), allow (outgoing), disabled (routed)

To	Action	From
--	-----	----
22/tcp	ALLOW IN	<MY_LOCAL_LAPTOP_IP_HERE>
80/tcp	ALLOW IN	Anywhere
8000/tcp	ALLOW IN	Anywhere
80/tcp (v6)	ALLOW IN	Anywhere (v6)
8000/tcp (v6)	ALLOW IN	Anywhere (v6)

```

azureuser@prasad-vm: ~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN 205.254.169.218
80/tcp ALLOW IN Anywhere
8000/tcp ALLOW IN Anywhere
22/tcp ALLOW IN Anywhere
80/tcp (v6) ALLOW IN Anywhere (v6)
8000/tcp (v6) ALLOW IN Anywhere (v6)
22/tcp (v6) ALLOW IN Anywhere (v6)

azureuser@prasad-vm: ~$
azureuser@prasad-vm: ~$

```